

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-6

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

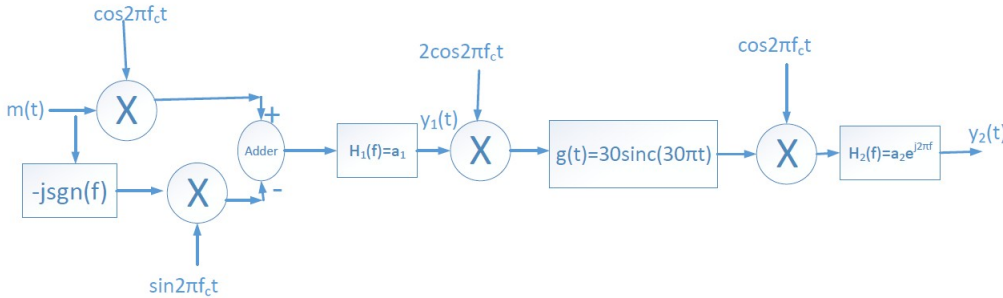
Date: 17-09-2025

Objectives

- SSB, FM

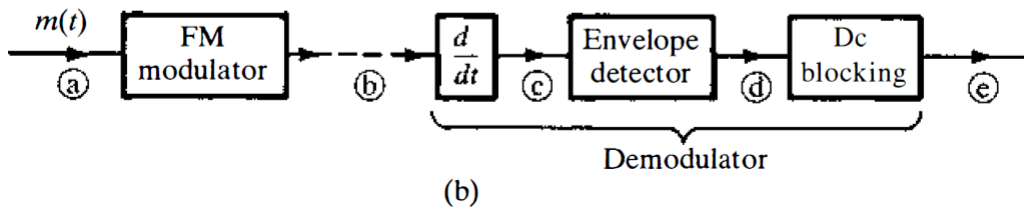
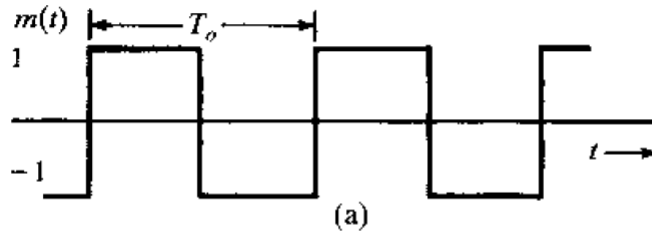
Tutorial Problems

1. A message signal $m(t)$ of bandwidth 10 Hz uses first SSB modulation over a channel of gain a_1 , then DSB-SC for a channel of gain a_2 . Find the output $y_2(t)$ in terms of the input signal $m(t)$.



2. A message signal $m(t) = 2B \text{sinc}(2B\pi t)$ frequency modulates a carrier signal $A \cos(2\pi f_c t)$ with frequency sensitivity factor k_f Hz/Volt.
 - (a) Find the integral equation of the FM signal.
 - (b) Find the BW of FM signal using Carson's rule.
3. A message signal $m(t)$ of bandwidth B phase modulates a carrier $c(t) = A \cos(2\pi f_c t)$ with a phase deviation constant k_p rad/volt. The PM modulated signal $x(t)$ is passed through a bandpass filter having a center frequency f_c Hz and bandwidth $4B$. Find the time-domain signal $y(t)$ at the output of the bandpass filter.
4. TB 5.4.2

- 5.4-2** A periodic square wave $m(t)$ (Fig. P5.4-2a) frequency-modulates a carrier of frequency $f_c = 10$ kHz with $\Delta f = 1$ kHz. The carrier amplitude is A . The resulting FM signal is demodulated, as shown in Fig. P5.4-2b by the method discussed in Sec. 5.4 (Fig. 5.12). Sketch the waveforms at points b , c , d , and e .



Solution

1. $y_1(t) = a_1(m(t) \cos 2\pi f_c t - m_h(t) \sin 2\pi f_c t)$
 $y_2(t) = a_1 a_2 (m(t-1) \cos 2\pi f_c (t-1))$
2. (a) $x(t) = A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau)$
 (b) $\text{BW} = 2(\Delta f + f_m) = 2(k_f m_p + B) = 2(2Bk_f + B)$
3. Solution is appended:

$$\begin{aligned}
 x(t) &= A \cos[2\pi f_c t + k_p m(t)] \\
 x(t) &= A \operatorname{Re} \left[e^{j2\pi f_c t + j k_p m(t)} \right] \\
 &= A \operatorname{Re} \left[e^{j k_p m(t)} e^{j2\pi f_c t} \right] \\
 &= A \operatorname{Re} \left[\left(1 + j k_p m(t) - \frac{k_p^2 m^2(t)}{2} - \frac{j k_p^3 m^3(t)}{6} + \dots \right) (\cos 2\pi f_c t + j \sin 2\pi f_c t) \right] \\
 x(t) &= A \cos 2\pi f_c t - A k_p m(t) \sin 2\pi f_c t \\
 &\quad - \frac{k_p^2 m^2(t)}{2} \cos 2\pi f_c t + \frac{k_p^3 m^3(t)}{6} \sin 2\pi f_c t + \dots
 \end{aligned}$$

First signal $\rightarrow$ An impulse in frequency domain
 Second signal $\rightarrow$ DSB-SC at f_c with BW $2B$
 Third signal $\rightarrow$ DSB-SC at f_c with BW $4B$
 Fourth signal $\rightarrow$ DSB-SC at f_c with BW $6B$
 and so on.

Thus $y(t) = \left[A \cos 2\pi f_c t - A k_p m(t) \sin 2\pi f_c t - \frac{k_p^2 m^2(t)}{2} \cos 2\pi f_c t + \frac{k_p^3 m^3(t)}{6} \sin 2\pi f_c t + \dots \right]$

4. Solution is appended:

4.4-2 Given that $f_c = 10$ kHz, $\Delta f = 1$ kHz, and the message is periodic square wave of period T_0 , the resulting FM signal simply switches instantaneous frequency from 11 kHz to 9 kHz and back over one period. Thus,

$$\varphi_{\text{FM}}(t) = A \cos [20000\pi t \pm 2000\pi t]$$

over any given half period. Now, after the ideal differentiator,

$$\dot{\varphi}_{\text{FM}}(t) = -(20000\pi \pm 2,000\pi)A \sin [2,000\pi t \pm 2000\pi t]$$

Next, after the envelope detector, the output will be a periodic square wave proportional to $(20000\pi \pm 2,000\pi)A$, with a dc offset. After dc blocking, the result is a periodic square wave proportional to $m(t)$. This is illustrated in Fig. S4.4-2.

